

USER REQUIREMENTS DOCUMENT

Surgical & Optical Supplies — Inventory Transaction Visibility

ClearView Medical Supplies | Internal Analytics Project

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Stakeholder	Naledi Mokoena, Head of Procurement & Inventory
Status	Under Review

1. Background & Storyline

ClearView Medical Supplies is a regional distributor of surgical instruments and optical products, serving hospitals, clinics, and optometry practices across four locations: North, South, East, and West. Their product catalogue spans four main categories: surgical tools (SURG), optical lenses (LENS), frames (FRM), and solutions (SOL), sold through a mix of Opticians and Surgeons across each branch.

In early 2026, Naledi Mokoena, Head of Procurement and Inventory, flagged a growing operational problem during a quarterly review: the company's inventory transaction records are being logged but never properly analysed. Orders come in, quantities are recorded, and risk levels are tagged — but no one is regularly checking whether high-risk orders are being fulfilled on time, which SKUs are moving fastest, or whether certain locations are consistently over-ordering.

The most pressing complaint from clinic administrators is stockouts of SURG-P-02 surgical packs, which carry a 30-day lead time and a Medium Risk tag. By the time the procurement team notices a shortage, it is already too late to reorder without disrupting scheduled procedures. Naledi put it plainly: "We have the data. We are just not reading it."

This document captures the requirements gathered from that initial stakeholder conversation and maps them to the Inventory_Transactions dataset as the foundation for a Week 1 SQL mini project.

2. Objective

Design and deliver an Inventory Transaction Visibility Report that enables procurement, branch managers, and clinical staff at ClearView Medical Supplies to monitor order volumes, identify high-risk and high-lead-time transactions, and surface which SKUs and locations require immediate stock management attention.

The project will use the Inventory_Transactions dataset (1,415 transaction records, 9 columns, covering January to September 2025) as its foundation, applying SQL-based analysis using SELECT, WHERE, ORDER BY, and LIMIT.

3. Problems Identified

The following pain points were identified during the stakeholder interview with Naledi Mokoena:

#	Problem	Business Impact
1	No consolidated view of high-risk transactions	192 High Risk transactions exist in the dataset but are not being tracked or escalated — procurement only reacts after a stockout occurs
2	Long lead times not factored into reorder planning	High Risk SKUs carry a 45-day lead time on average. By the time a shortage is noticed, delivery is over a month away
3	No visibility into which SKUs are moving fastest	LENS, FRM, SURG, and SOL categories have very different order volumes and unit quantities — procurement treats them the same
4	Location imbalances not monitored	South has the most transactions (383) while North has the fewest (333). Uneven demand is not being used to guide stock allocation
5	Staff order patterns not reviewed	Only 4 staff members (Optician_A, Optician_B, Optician_C, Surgeon_K) process all orders — unusual spikes by individual are not flagged
6	Revenue figures not linked to inventory risk	High-value orders (up to R3,750 per transaction) often correspond to high lead time SKUs — no process exists to prioritise these

4. Target Audience

The following roles are the primary and secondary users of this report:

Role	Type	Primary Need
Head of Procurement (Naledi)	Primary	Overall view of risk exposure and lead time gaps across all locations
Branch Managers (x4 locations)	Primary	Which SKUs at my location are High Risk and need urgent attention?
Opticians & Surgeons (ordering staff)	Secondary	Confirmation that items ordered have acceptable lead times before committing to a patient
Finance Team	Secondary	Understand revenue concentration by SKU and risk tier
Data Analytics Intern (you)	Builder	Produce clean, query-ready data and documented findings using Week 1 SQL skills

5. Use Cases

UC-01: View All High-Risk Transactions

User Story	As a branch manager, I want to see all transactions tagged as High Risk so that I can escalate reorders before they become stockouts.
Acceptance Criteria	Query returns order_id, date, SKU, quantity_sold, location, lead_time, and risk_level. Filters to risk_level = 'High Risk' only. Results are ordered by lead_time descending so the most urgent items appear first.
SQL Concepts Used	SELECT, WHERE, ORDER BY

UC-02: Find the 10 Largest Orders by Quantity

User Story	As a procurement officer, I want to see the 10 highest-quantity transactions so that I can identify which orders are placing the most strain on available stock.
Acceptance Criteria	Query returns order_id, SKU, quantity_sold, location, staff_id, and revenue. Results are ordered by quantity_sold descending. Only the top 10 rows are returned.

SQL Concepts Used	SELECT, ORDER BY, LIMIT
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UC-03: Filter Transactions by Location and Risk Level

User Story	As a branch manager in the South location, I want to see all Medium and High Risk transactions at my branch so that I can review which orders to prioritise this week.
Acceptance Criteria	Query filters on location = 'South' AND risk_level in ('Medium Risk', 'High Risk'). Returns order_id, date, SKU, quantity_sold, lead_time, and risk_level. Ordered by risk_level then lead_time descending.
SQL Concepts Used	SELECT, WHERE (AND / OR), ORDER BY

UC-04: Identify High-Revenue Orders with Long Lead Times

User Story	As the Head of Procurement, I want to identify orders with revenue above R500 and a lead time of 30 days or more, so that I can ensure high-value stock is prioritised in our replenishment schedule.
Acceptance Criteria	Query applies WHERE revenue > 500 AND lead_time >= 30. Returns order_id, date, SKU, location, revenue, lead_time, and risk_level. Ordered by revenue descending.
SQL Concepts Used	SELECT, WHERE (AND), ORDER BY

6. Success Criteria

The project will be considered successful when all of the following conditions are met:

- All four use cases have working SQL queries that return correct and reproducible results.
- The dataset has been explored and a data quality report documents any issues found across all 9 columns.
- The semicolon delimiter and date format (M/D/YYYY) have been acknowledged and handled correctly in any tooling used.
- Findings are recorded in a structured analysis log with the query used, a summary of the output, and a plain-language interpretation for each use case.
- Naledi Mokoena (mock stakeholder) can identify all High Risk transactions and the top revenue-at-risk orders without opening a raw CSV file.
- All work is reproducible: queries, cleaning steps, and findings are stored in a documented project folder.

7. Data Needed

The dataset used is: Inventory_Transactions.csv (1,415 rows, 9 columns, January to September 2025). Note: the file uses a semicolon (;) as its delimiter, not a comma.

The following columns are relevant to the inventory visibility use cases:

Column	Data Type	Role in Analysis
order_id	String	Unique transaction identifier (e.g. ORD-501)
date	String (M/D/YYYY)	Transaction date — note non-standard format, needs handling
SKU	String	Product code; prefix indicates category (SURG, LENS, FRM, SOL)
quantity_sold	Integer	Units ordered per transaction — demand signal
location	String	Branch location (North / South / East / West)
staff_id	String	Ordering staff member — role encoded in prefix (Optician / Surgeon)
revenue	Float	Transaction revenue value — ranges from R4.25 to R3,750
lead_time	Integer	Days from order to delivery (7, 14, 30, or 45 days)
risk_level	String	Inventory risk classification: Low Risk / Medium Risk / High Risk

Note: A stock_available or reorder_point column is not present in this dataset. Risk_level serves as the proxy indicator for inventory urgency. High Risk transactions with lead_time = 45 days should be treated as requiring immediate escalation.

8. Data Quality Checks

The following checks must be completed before any analysis is performed. Results should be recorded in the project data quality log.

#	Check	Why It Matters	Expected SQL Approach
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1	Null / missing values in all columns	Initial exploration shows 0 nulls — this should be verified formally before any filtering is applied	WHERE column IS NULL
2	Duplicate order_id values	Each order_id should appear exactly once. Duplicates would double-count quantities and revenue	COUNT(*) vs COUNT(DISTINCT order_id)
3	Valid risk_level values only	Only 'Low Risk', 'Medium Risk', and 'High Risk' are expected — any other value is a data entry error	SELECT DISTINCT risk_level
4	Lead time restricted to expected values	Lead times appear to be categorical (7, 14, 30, 45 days). Any value outside this set is suspicious	WHERE lead_time NOT IN (7, 14, 30, 45)
5	Quantity sold in valid range	Min = 1, Max = 25 in current data — check for zero or negative values that would indicate data entry errors	WHERE quantity_sold <= 0
6	Revenue non-negative and non-zero	A revenue of 0 or below would indicate a missing price or a return that was not properly categorised	WHERE revenue <= 0
7	Date format consistency	Date is stored as M/D/YYYY (non-standard). Verify all values match this pattern before any date-based ordering or filtering	WHERE date NOT LIKE '%/%/%'
8	SKU prefix consistency	All SKUs should begin with SURG, LENS, FRM, or SOL. Flag any values outside these four prefixes	SELECT DISTINCT SKU — check prefix manually
9	Staff ID format consistency	All staff_id values follow Optician_X or Surgeon_X format. Flag any values that deviate	SELECT DISTINCT staff_id ORDER BY staff_id
10	Risk level aligned with lead time	High Risk always maps to lead_time = 45 in this dataset. Verify this holds — any mismatch is a classification error	WHERE risk_level = 'High Risk' AND lead_time != 45

9. Next Steps

With this requirements document approved, the following project stages are ready to begin:

1. Data Collection — confirmed (dataset loaded, 1,415 rows, 9 columns)
2. Data Exploration — initial schema, value ranges, and category distributions completed as part of this document
3. Data Cleaning — run all 10 quality checks above; address semicolon delimiter and date format in tooling
4. Data Testing — validate cleaning output against original row counts and distinct value sets
5. Analysis — write and run SQL queries for all four use cases
6. Findings — document results and present to mock stakeholder (Naledi)